

Individual Assessment Coversheet

To be attached to the front of the assessment.

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Group: 4

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Indicate	Yes	No
Plagiarism report attached		

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Signature 	Date 29 August 2024
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Lecturer's Comments:

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Section A

Question 1

1.1) A transaction processing system (TPS) refers to a system that supports business processes related with forms of transaction including billing, employee payroll as well as accounting. Dion can easily transition to TPS systems, because he can perform all relevant invoices, grant writing and business development on the same system instead of having to alternate between Microsoft Word and manual processes. This will greatly increase the overall efficiency and effectiveness of D5 Consulting when performing any form of business processes.

An online transaction processing (OLTP) system is a subset of TPS that processes transactions immediately without accumulating it into a batch. This can be implemented by Dion to automatically process the numerous invoices required by his business's clients. This will remove the necessity for Dion to spend his time tracking these invoices and can focus his attention on other business processes as well as lighten his workload. Additionally, this increase in efficiency can help improve the relationship between D5 Consulting and its customers, because client work can now be completed in a timely manner.

A batch processing system is another subset of TPS where several transactional processes are accumulated processed simultaneously after a period of time. This processing system can be applied to automate the transaction process between D5 Consulting and its contractors. This can be done by accumulating the required payment details of all contractors and then processing all these transaction at the same time when payment is required. Therefore, this system will prevent Dion from paying his contractors late again, enables him to easily track these payment details as all these details are stored in the same location and further lightens his workload.

1.2) The implementation of an invoicing system designed to automate the various business processes of D5 Consulting, such as contractor payments and client invoices, can introduce benefits to organisation.

An automated invoicing system can help to enhance the business's cash and financial management as well as reduce costs, because Dion can use these systems to automatically track and analyse all payment occurrences and make him less reliant on external contractors, enabling him to save on labour costs by hiring less contractors. Additionally, these invoicing systems allows D5 Consulting to complete their invoices in a timely manner as it can be processed immediately without having to wait for a contractor to complete their tasks. Furthermore, these invoicing systems can also improve accuracy as the invoices are done by a program, which prevents any instances of human error or misunderstanding.

Occurrences of automated invoicing systems in situations outside of D5 Consulting can include the implementation of these systems in the town of Vegreville. This town applied an automated invoicing system to simplify the process of pet owners obtaining dog tags without needing to come to the office. This is done by making the system send a notice of renewal to the owners and allowing them to pay through the invoicing system. This system helps to save time for both the customers and the organisation, however there are issues associated with this system as some individuals forget to respond to the notice sent to them (Simpson, 2012). Furthermore, Equifax

Canada is also implementing an automated invoice system in order to streamline multiple transaction processes in their global billing system (Delhi, 2020). This provides the benefit of increased efficiency due to its ability to manage multiple processes, however it will require individuals with expertise to ensure that it remains functional due to its complexity.

Question 2

2.1) Traditional retailers, if they are unable to adapt and remain competitive, can be negatively impacted by the competition provided by international players. Consumers are more likely choose to interact with international players instead of local retailers, because they provide a more convenient experience for consumers to locate and acquire goods and services. Therefore, local retailers will gradually lose potential customers, which will lead to reduced profit and the potential for some retailers to be unable to conduct business anymore. Moreover, international players can also make it more difficult for new retailers to enter the industry depending on the popularity of their business in certain communities.

Additionally, there are other effects of competition between companies that includes an increased incentive for innovation, the need to better the quality of services and products and the lowering of service and product prices (Stucke, 2013). Local retailers can take these effects into consideration when determining strategies to remain competitive with international players and other retailers. An effective strategy is to implement their own e-commerce platform due to the popularity of the e-commerce industry and the barriers of entry being low.

An e-commerce platform can prove to be beneficial to retailers as it enables them to reach new consumers who were unaware of their place of business. Furthermore, the retailer can innovate within the e-commerce industry, due to the lack of innovations, enabling them to grab the attention of consumers and increase their efficiency and effectiveness. Moreover, retailers can also lower the prices of their products and services to raise the number of potential consumers and increase the potential for profit.

2.2) One of the main objectives of a business that makes use of an e-commerce environment is to ensure that the consumer experience is positive, which can lead to a greater potential for profit. The reason for this increase in potential profit is found in the different factors that contribute to a positive customer experience like the quality of the e-commerce platform.

Guaranteeing that the quality of the service that the consumers experience on the e-commerce platform is excellent, will help to improve the consumer's perspective and relationship with the organisation and can lead to long-term benefits such as profit and customer loyalty. Customer loyalty can further increase potential profit, because consumer will be more likely to conduct business with the company multiple times and suggest to the company's services and products to other individuals (Sukendi, 2021).

Moreover, personalisation can also improve the consumer experience by providing a specialised experience for each individual customer. Strategies introduced by personalisation include using data from users to recommend and offer discounts to appropriate goods and services as well as create advertisements based on consumer behaviour (Athena Information Solutions Pvt. Ltd, Mumbai, 2024). These tactics makes it more likely for consumers to purchase new products and services, because the content provided is shown to be relevant to each individual customer.

Personalisation will improve the overall consumer experience by providing a more convenient experience and will have other long-term effects such as customer satisfaction and loyalty.

Question 3

3.1) The implementation of the Dataportal enables Airbnb to store all its data resources in a central location. By storing the data in a central location, it increases the efficiency of employees when they search for specific pieces of data as all the information they need can be found on Dataportal and they will no longer have to search through Airbnb's other tools. In scenarios where employees need to search for information related to a specific customer to solve queries, it can easily be found as all forms of data associated with that customer can be searched for and accessed simultaneously.

Additionally, Dataportal provides extra details that describes the background of a piece of data such any links it has to other pieces of data, which can improve data management. Employees will be able to easily identify the differences between similar data sources enabling them to determine the correct information that they are looking for. This feature offered by Dataportal can prove to be beneficial for employees, because it makes it possible for them to search for and identify information of a certain customer that has details, like their name and surname, that are the same as other customers by pinpointing the more unique and identifiable background details.

Furthermore, these extra details that are offered by the Dataportal can also introduce new insights and allow the employees to analyse the data. This improvement of the ability to analyse the data in greater detail improves the decision-making of employees as they can fully inform themselves on the context of each piece of data. A scenario of the application of this benefit includes employees determining if a customer justifies a refund after their experience with a rental home by identifying similar complaints made towards the same rental home and other homes that are owned by the same individual.

3.2) Data science refers to the process of gathering and analysing data to provide new insights on situations. Data science can be beneficial to customers of Airbnb in a variety of manners such as improved searching of rental homes for customers.

Data science improved the search function of Airbnb by offering rental homes in a specific region to customers that were booked more often when compared to homes within the surrounding region. This increases customer satisfaction, because it grants a higher likelihood for the offered rental housing to be of high quality due to its popularity. Furthermore, service personalisation also benefits in this scenario as the search function can also take into consideration the customer's previous behaviour to offer popular rental homes that are more appropriate for the customer.

Moreover, data science can also be utilised by the employees of Airbnb to make informed decisions that will improve the overall customer experience with the organisation. Making use of data science techniques to gather information from customer behaviour to determine if any Airbnb processes requires improvements. This can improve customer experience, because various Airbnb systems can be redesigned to make the whole process more convenient for customers. Additionally, these changes made towards Airbnb systems can improve service personalisation by introducing personalisation features for customers to utilise and further improve their experience with Airbnb.

3.3) The implementation of data science in different Airbnb systems introduces new job opportunities for individuals who want to work for Airbnb. By adding the role of data scientists to the workforce of the organisation, allows Airbnb to easily utilise diversity hiring practices to incorporate more diverse employees into their workplace. This can increase the productivity and performance of the organisation, because these individuals will have the skills to fully analyse all aspects of business data and provide informed strategies that can increase the effectiveness and efficiency of Airbnb systems and processes.

Furthermore, data science can also be used to develop or update already existing diversity hiring practices to increase effectiveness. This can be done by analysing how other organisation hiring practices are able to effortlessly attract talented and diverse individuals to their workplace and Airbnb can then implement these findings to improve their own hiring systems and practices. In addition, this can improve overall business efficiency and performance as the new hires will come equipped with the necessary skills to perform their tasks effectively.

However, when using data science in this and other scenarios, caution must be taken as the data the business analyses could be biased or misleading. The integrity of the data must be investigated, because if it is not good quality data then incorrect decisions can be made that can harm the overall performance of Airbnb. In this scenario, Airbnb could implement new diversity hiring practices that can attract diverse individuals who are not well equipped for their positions within the business leading to the potential decrease in performance.

Question 4

4.1) Cloud-based services refers to services that offers IT resources over the internet with a pricing model where you pay-as-you-go. Software as a Service (SaaS) is an example of a cloud-based service where the organisation has access to applications over the internet that have introduced various benefits to the roles of IT departments. The roles and responsibilities of the IT department that have been impacted by SaaS are administrative tasks, maintaining organisation infrastructure and resources, communication, technical support as well as completing work related objectives.

The SaaS adoption has improved the communication process for the IT department when interacting with themselves and others within the organisation. Email services like Gmail from Google and Outlook from Microsoft are examples of SaaS services that establishes effective communication, because users can easily send messages to the recipient if they are unable to easily interact with them and it allows them to attach any additional resources like documents if required. Another advantage of these services is that information that is sent can be viewed again at a later date preventing the individuals from the IT department to having to resend information.

Furthermore, the SaaS model allows the IT department to manage and maintain the infrastructure within the organisation less. This is possible, because the infrastructure that supports the applications in the SaaS model are maintained by another organisation, like Google, that manages that software application. This allows other organisations to spend less on their own infrastructure and the IT department can focus more on their other responsibilities and tasks.

In addition, the integration of the SaaS model allows individuals in the IT department share resources with each other as well as easily collaborate and complete work-related tasks on a

shared software application. Google Docs is an example of a SaaS model that provides this advantage by allowing multiple users can add their contribution on their own time or simultaneously.

However, the SaaS model introduces additional administrative tasks that the IT department will need to take care of. These administrative tasks include managing user accounts for the SaaS model and ensuring that all these accounts have the appropriate licenses to use the software applications.

Moreover, the IT department may not be able to provide adequate technical support to individuals within the organisation if they experience issues with the SaaS model. They will need to rely on the organisation that manages the SaaS model to provide technical support that can solve any issues that they are experiencing.

4.2) Performing remote work within organisations is only possible due to the main benefits provided by the software as a service cloud computing model, which is its scalability as well as its overall accessibility. The software application can be accessed by employees at anytime and anywhere as long as they have a stable internet connection, and organisations are also able to effortlessly scale the number of resources and users associated with the application software.

Furthermore, the number of organisations that uses cloud services that are provided by cloud computing companies such as Google and Microsoft to perform work remotely increased due the lockdowns occurring during the Covid-19 pandemic. During this period of time, many organisations relied on software applications from cloud providers as employees were unable to work on-premises. When utilising the software as a service cloud computing model, the organisations discovered that they could save costs by downsizing branch locations that has its own wide area network or virtual private network, since their employees preferred the consistent look and feel that the application software provided whether in or out of the office (Nolle, 2022).

Microsoft has created a package called Microsoft 365 where it contains various software applications that can be accessed over the internet for remote work and these applications include the Teams application, where employees are able to easily conduct meetings, send work related resources like documents and organise groups where employees are able to send messages to each other. Additionally, Microsoft 365 also comes packaged with the Word, Excel and PowerPoint software applications that provides employees with resources to create professional documents, spreadsheets and presentations and it enables them to download the applications to perform the work without an internet connection.

Moreover, Google also provides software applications that enables remote work like Google Docs where employees are able to work on the same document simultaneously allowing for an increase in efficiency and for the employees to review each other's work. In addition, Google Drive enables the organisation to store data with scalable storage and any employee will have access to the data at any time.

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